Unified Detection of Tenant-Isolation Violations and QoS Fairness Breaches in Multi-Tenant Cloud Platforms

Course Project for IT418 (Cloud Computing) Course

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Base Paper

- *Title:* Dynamic Application Security Testing for Kubernetes Deployment: An Experience Report from Industry
- *Authors:* Shazibul Islam Shamim, Hanyang Hu, Akond Rahman
- *Venue:* FSE '25 (33rd ACM Intl. Conference on the Foundations of Software Engineering), Trondheim, Norway, June 2025
- *DOI:* 10.1145/3696630.3728573

What the Base Paper Does?

- Industry experience report on using DAST (Dynamic Application Security Testing) for Kubernetes configuration security
- Partnered with "Company-Z"; evaluated 3 DAST tools on a live K8s environment
- Found thousands of CIS-recommended configuration violations
- *Recommendation:* DAST should complement static analysis — static checks miss violations that only appear at runtime
- Treats Kubernetes as a generic deployment target, not as a multi-tenant cloud substrate

Limitations of the Base Paper

- No implemented artifact, findings/recommendations only, no tool built
- Not framed around cloud computing's core concept, ignores multi-tenancy & resource pooling
- Single-dimension notion of "violation", security only; ignores fairness/QoS
- No quantitative evaluation, no precision, recall, FP rate, or latency
- Single signal source, relies only on control-plane (API/audit) signals

Our Novelty - What We Address

- We directly address Limitations 1, 2, 3, 4, and 5
- *Core novelty*: unify two failure modes of shared cloud infrastructure into one pipeline:
- Isolation violations (security), Tenant A improperly accesses Tenant B's resources
- Fairness/QoS violations (resource management), Tenant A's workload starves Tenant B's promised performance ("noisy neighbor" problem)
- Reframes the base paper's generic recommendation around cloud computing's defining property: multi-tenant resource pooling

Summary / Takeaway

- *Base paper*: generic, security-only, no artifact, no evaluation
- *Our contribution*: a working, evaluated system that puts cloud computing's defining property — safe, fair multi-tenant resource sharing — at the center
- One monitoring investment, two failure modes covered: isolation and fairness

THANK YOU